DO NOW — RECALL

Lesson 063 · Thinking Logically: Conditions and Decision

Name: _____ Class: _____ Date: _____

1. What does a Boolean expression evaluate to?

2. What is a decision boundary?

3. Why is condition order important in else-if?

4. Simplify (A AND B) OR (A AND NOT B), showing the Boolean laws used.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

True or False.

2. Model answer

A value at which condition outcome changes.

3. Model answer

Earlier branch can prevent later branch ever being checked.

4. Model answer

A AND (B OR NOT B) = A AND 1 = A. Use distribution, complement and identity.

By the end, you can…

• Identify decisions in the solution and write precise Boolean conditions, including valid-data boundaries.

• Order conditions so cases are mutually understood: mark>=70 must be considered before mark>=40.

• Use truth tables or decision tables to test normal, erroneous and boundary input values.

Vocabulary — know these terms

predicate — Boolean condition.

selection — Branch based on a condition.

boundary test — Input near comparison threshold.

short-circuit — Skipping redundant Boolean evaluation.

NEW LEARNING

Read, annotate and explain

Identify

Identify decisions in the solution and write precise Boolean conditions, including valid-data boundaries.

Model

Order conditions so cases are mutually understood: mark>=70 must be considered before mark>=40.

Evaluate

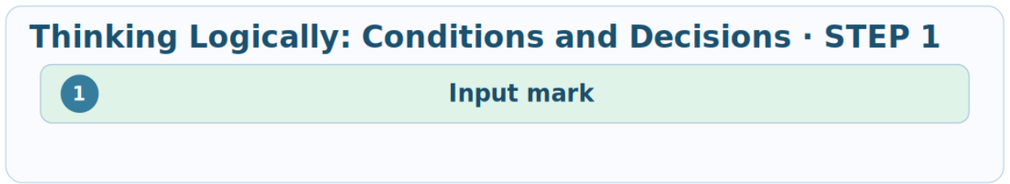
Use truth tables or decision tables to test normal, erroneous and boundary input values.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

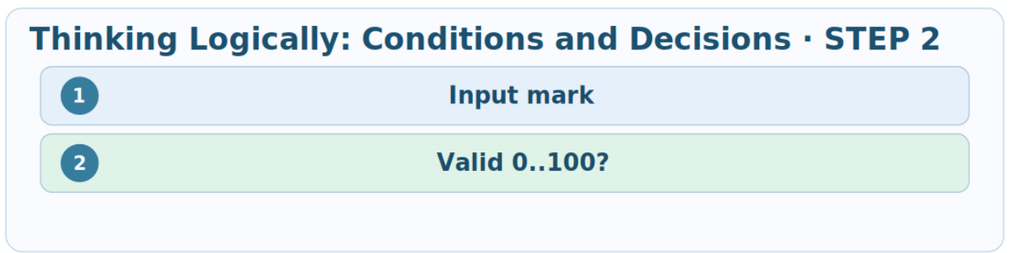
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



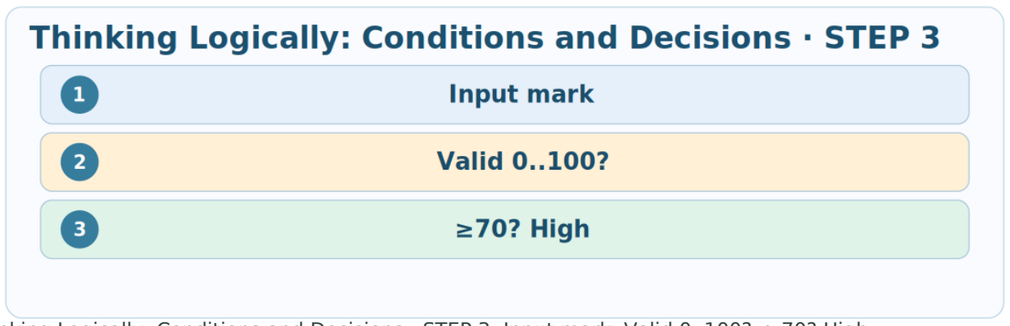
Thinking Logically: Conditions and Decisions · STEP 1: Input mark

New Learning



Thinking Logically: Conditions and Decisions · STEP 2: Input mark; Valid 0..100?

New Learning



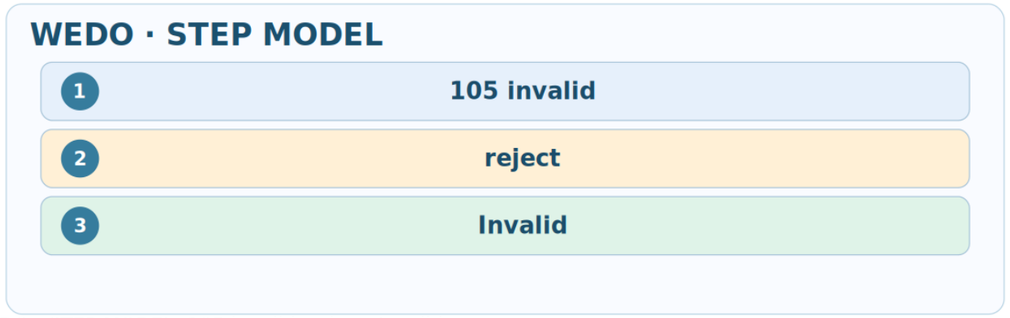
Thinking Logically: Conditions and Decisions · STEP 3: Input mark; Valid 0..100?; ≥70? High

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace Grade(105). Why should validity be checked first?

Without range check, 105 would incorrectly be classified High.



WEDO · STEP MODEL: 105 invalid; reject; Invalid

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

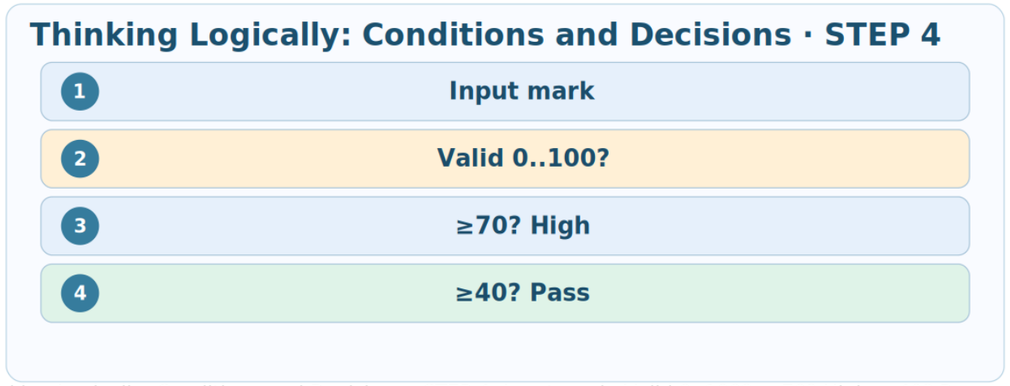
Guided 1 — CHECK

Explain the decision
Without range check, 105 would incorrectly be classified High.

New Learning · original lesson diagrams

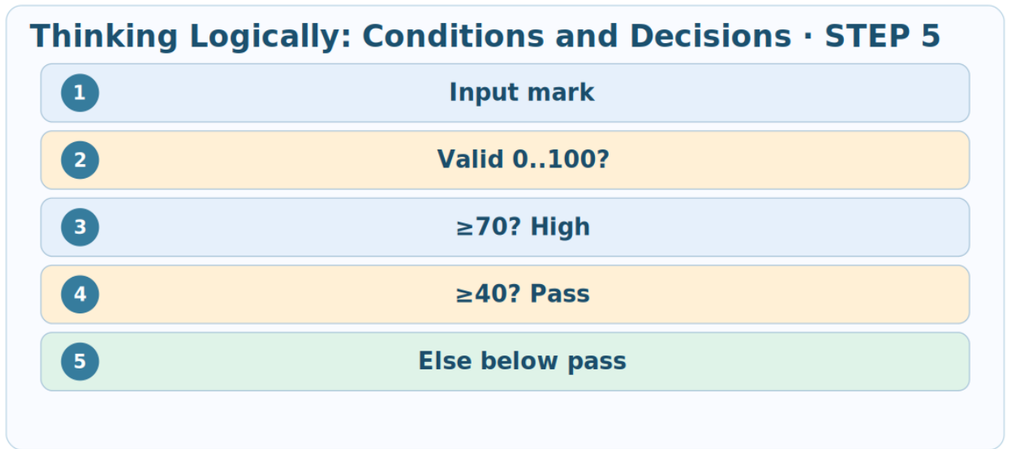
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Thinking Logically: Conditions and Decisions · STEP 4: Input mark; Valid 0..100?; ≥70? High; ≥40? Pass

New Learning



Thinking Logically: Conditions and Decisions · STEP 5: Input mark; Valid 0..100?; ≥70? High; ≥40? Pass; Else below pass

I DO — WORKED EXAMPLE

Follow the worked method

Trace Grade(40) using the conditions shown.

Explain the decision

Check comparisons in order; stop after first matching return.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

With else-if, a later condition always runs even if an earlier branch matches.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

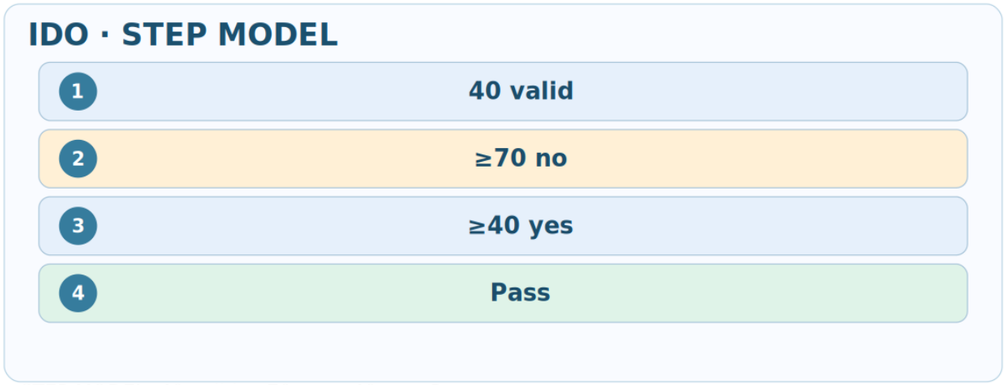
A program checks if(mark>=40) before else-if(mark>=70), then outputs Pass for 85. Explain bug and repair. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace Grade(40) using the conditions shown.

Check comparisons in order; stop after first matching return.



IDO · STEP MODEL: 40 valid; ≥70 no; ≥40 yes; Pass

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Trace Grade(105). Why should validity be checked first?
Source code / input:
105>100, so return Invalid before the threshold branches.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace Grade for marks 39,40,69,70 and give output at each. [4]

Question 2

Write a Boolean rule for a valid age 11–18 inclusive, and give two boundary tests. [6]